

Classifying Single-Trial EEG During Motor Imagery by Iterative Spatio-Spectral Patterns Learning (ISSPL)

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Abstract—In most current motor-imagery-based brain-computer interfaces (BCIs), machine learning is carried out in two consecutive stages: feature extraction and feature classification. Feature extraction has focused on automatic learning of spatial filters, with little or no attention being paid to optimization of parameters for temporal filters that still require time-consuming, *ad hoc* manual tuning. In this paper, we present a new algorithm termed iterative spatio-spectral patterns learning (ISSPL) that employs statistical learning theory to perform automatic learning of spatio-spectral filters. In ISSPL, spectral filters and the classifier are simultaneously parameterized for optimization to achieve good generalization performance. A detailed derivation and theoretical analysis of ISSPL are given. Experimental results on two datasets show that the proposed algorithm can correctly identify the discriminative frequency bands, demonstrating the algorithm's superiority over contemporary approaches in classification performance.

Index Terms—Brain-computer interface (BCI), common spatial patterns (CSPs), maximal margin hyperplane, single-trial EEG, motor imagery, spatio-spectral filters.

I. INTRODUCTION

A BRAIN-COMPUTER interface (BCI) provides a direct communication pathway between a human and an external device. With recent advances in technology and knowledge, researchers can now conceivably attempt to produce BCIs that augment human functions rather than simply restoring them, previously only the realm of science fiction [1]. In the study of noninvasive EEG-based BCIs, of particular interest is the motor-imagery-based BCI [2]. By imagining movements of different parts of the body, trained subjects can voluntarily regulate their μ or β rhythms over sensorimotor cortices [3]. This type of BCI system has become an active research theme due to its relatively robust communication performance and its potential neurophysiological significance for studying the underlying mechanism of motor imagery. A big challenge, however, is for this BCI

to correctly and efficiently identify and extract subject-specific features from the blurred scalp EEG and translate those features into device commands. Signal processing and machine learning algorithms have been devised to address this issue [4]–[6]. In general, these algorithms focus on identifying discriminative patterns in the spatial and frequency domain.

A highly successful spatial filtering algorithm, as evidenced by BCI Competition II and III [5], [7], is *common spatial patterns* (CSPs) (also known in the machine learning field as the *Fukunaga-Koontz transform*), which was first proposed as an application of the *Karhunen-Loève transform* for classification by Fukunaga *et al.* [8], and was later introduced for use in BCI systems by Ramoser *et al.* [9] (see [30] for a recent review on CSP). To a great extent, CSP's success in the motor-imagery-based BCIs can be attributed to the underlying neurophysiological phenomena accompanying motor imagery, termed *event-related desynchronization* (ERD) and *event-related synchronization* (ERS) [10]. Since the CSP algorithm is designed to find spatial projections that can maximize the power ratio of two classes, it is able to capture the optimal spatial projections that account for ERD/ERS. Despite its prevalent use for feature extraction in motor-imagery-based BCIs, the original CSP algorithm only searches for optimal projections in the spatial domain. In contrast, the "optimal" temporal filter is often found manually by comparing the power spectra of different classes, or is simply set to a broad range that includes both μ and β rhythms.

Until now, several methods for modifying CSP to exploit temporal structures in EEG signals have been devised to determine subject-specific "optimal" temporal filters [11]–[13]. One such attempt is the *common spatial spectral patterns* (CSSPs) algorithm that simultaneously embedded a first-order finite impulse response (FIR) temporal filter into CSP for optimization [11]. However, first-order FIR filters are very limited in their flexibility to choose among different frequency bands. Dornhege *et al.* later proposed the *common sparse spectral spatial patterns* (CSSSPs) algorithm to increase the flexibility of FIR filters [12]. To avoid the potential overfitting problem, they incorporated a regularization term to sparsify the solution. Yet, CSSSP is computationally expensive, which is undesirable for online applications. Recently, Tomioka *et al.* proposed to optimize temporal filters equivalently in the frequency domain (which we shall call *spectral filters* henceforth) [13]. The original CSP algorithm was alternated with a procedure for optimizing spectral filters, in which the square root of the Rayleigh quotient was adopted as the objective function. Motivated by this idea as

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attempting to optimize spectral filters, in this paper, we present a new approach to further improve the classification performance of the motor-imagery-based BCIs. In the proposed framework, the optimization of spectral filters is directly related to the classification of different classes, which can be demonstrated to have good generalization performance according to statistical learning theory. Moreover, no particular statistical information is assumed for deriving the approach.

The main contributions of this paper include the following.

- 1) We review the problem of single-trial motor imagery classification (Section II-A) and present the details of our proposed algorithm for optimizing spatio-spectral filters and the classifier, in which the idea of the maximal margin hyperplane is used (Section II-B).
- 2) We compare the proposed algorithm with the algorithm presented by Tomioka *et al.*, pointing out their distinctions (Section II-C).
- 3) We present some experimental results to validate the proposed algorithm and compare it with the state-of-the-arts algorithms to demonstrate its effectiveness (Section III).

We discuss several theoretical/implementation issues concerning the proposed algorithm and possible extensions to the current research in Section IV, and end the paper with concluding remarks.

II. METHODS

A. Problem Definition

In the binary classification case, the task is to assign single-trial EEG data to one of two predefined categories, e.g., “+” or “−”. To achieve this, the data are first sent to a feature extractor whose purpose is to exploit the structure of the data to obtain a discriminative feature vector in some subspace that encodes the user’s messages or commands. The feature vector is then passed to a classifier that evaluates the evidence presented and makes a final decision as to the assigned class.

In this paper, we are interested in extracting the *band power* (as its name suggests, the power in certain frequency bands) feature for classification. Previous neurophysiological studies show that motor imagerys are accompanied by a short-lasting decrease or increase of EEG power in specific frequency bands over the sensorimotor cortex [10]. As such, the feature extractor should maximize the discriminability of the band power features between two classes. A typical feature extractor for motor imagery BCI data consists of multiple pairs of linear spatiotemporal filters. In each pair of spatiotemporal filter, the temporal filter aims to capture useful information in the frequency domain, while the associated spatial filter extracts information by spatially projecting the multichannel data into a single dimension.

Assuming the feature extractor contains m pairs of linear spatiotemporal filters in total, we can represent the filtering operators by means of matrices or vectors. Specifically, denote the EEG data matrix of a single trial by $\mathbf{X} \in \mathbb{R}^{D \times T}$, where D is the number of channels and T is the number of discrete time points in a selected time window within the trial. If we assume periodic extensions of the finite length EEG signal at

the boundary, then, for the j th pair of filters, the temporal filter can be denoted by a circulant matrix $\mathbf{A}_j \in \mathbb{R}^{T \times T}$, with the k th diagonal of the matrix containing the k th filter coefficient, which wraps around to continue as the $(k - T)$ th diagonal [14]. Its associated spatial filter can be denoted by $\mathbf{s}_j \in \mathbb{R}^D$. The band power feature component after the j th filtering step can, therefore, be written in the following nonlinear form w.r.t. $\mathbf{X}$:

$$\begin{aligned} \phi_j(\mathbf{X}; \mathbf{A}_j, \mathbf{s}_j) &= \|\mathbf{s}_j^H \mathbf{X} \mathbf{A}_j\|_2^H \\ &= \mathbf{s}_j^H \mathbf{X} \mathbf{A}_j \mathbf{A}_j^H \mathbf{X}^H \mathbf{s}_j, \quad j = 1, \dots, m \end{aligned} \quad (1)$$

where the superscript H denotes Hermitian transposition, i.e., the operation of transposition combined with complex conjugation, which is the same as transposition in the real-valued case.¹ The feature vector ϕ is then formed by all the m feature components

$$\phi = [\phi_1(\mathbf{X}; \mathbf{A}_1, \mathbf{s}_1), \dots, \phi_m(\mathbf{X}; \mathbf{A}_m, \mathbf{s}_m)]^H. \quad (2)$$

The resultant feature vector ϕ needs to be classified by a pattern classifier. A linear classifier is often employed for the classification of the feature vector to guarantee good generalization ability and to avoid the potential overfitting problem typically suffered by nonlinear classifiers [5], [15]. In this case, the discriminant function of the classifier can be written as

$$o(\phi) = \mathbf{w}^H \phi + b \quad (3)$$

where $\mathbf{w} = [w_1, \dots, w_m]^H$ is the weight vector and b the bias.

Both the feature extractor and the classifier should be learned from empirical data (the training set $\mathcal{D}$). In conventional learning paradigms, learning of the feature extractor and the classifier are treated as two independent stages, in which although there are concepts from statistical learning theory characterizing the generalization performance of the classifier [16], feature extraction is done without specifically considering the generalization performance. When learning the feature extractor, spatial filters are often calculated by methods such as *common average referencing* (CAR), *surface Laplacian* (SL), or *source imaging techniques* [17], which are inspired by the neurophysiological basis of EEG [18], or popular statistical multichannel decomposition methods such as CSP, *principal component analysis* (PCA), and *independent component analysis* (ICA) [4], [19], [20], whereas temporal filters are generally determined approximately from *a priori* neurophysiological information or by time-consuming manual tuning based on the power spectra of some characteristic channels (e.g., the C3 and C4 channels when classifying left-hand and right-hand movements). Such methods of determining temporal filters should not be regarded as machine learning approaches due to their arbitrary nature.

B. New Approach for the Classification of Motor Imagery

We present the *iterative spatio-spectral patterns learning* (ISSPL) algorithm to automatically learn spatio-spectral filters and the classifier sequentially from labeled multichannel EEG data in an iterative fashion. In each iteration, the spatial filters are

¹For simplicity, we will adopt this notation for both complex- and real-valued cases throughout the paper.

firstly learned based on the spectral filters optimized in the preceding iteration. Then, both the spectral filters and the classifier are learned simultaneously by being parameterized via a maximal margin hyperplane. The alternating optimization procedure repeats until certain stopping criterion is satisfied (this will be described in detail). As soon as the spatio-spectral filters and the classifier are obtained by ISSPL, a test sample with unknown label can be readily classified into one of the two predefined categories.

1) *Learning Spatial Filters*: During each iteration, spatial filters are generated using the standard CSP algorithm. To maximize the discriminability of the band power feature, the algorithm makes use of the eigenstructure of the spatial covariance matrices to find low-dimensional linear subspaces in which the power ratio of two sets of data can be maximized.

To be more specific, given the training set $\mathcal{D}$ and a fixed temporal filter $\mathbf{A}_j$, denote the mean spatial covariance matrices for the two classes by $\mathbf{R}_{j+}$ and $\mathbf{R}_{j-}$, respectively, with

$$\mathbf{R}_{j+/-} = \langle \mathbf{X} \mathbf{A}_j \mathbf{A}_j^H \mathbf{X}^H \rangle_{+/-} \quad (4)$$

where $\langle \cdot \rangle_k$ denotes the ensemble averaging operator over multiple trials in $\mathcal{D}$ for class k . The CSP algorithm aims to find two sets of mutually orthogonal spatial projections that maximize the following two power ratios, respectively:

$$\frac{\mathbf{s}^H \mathbf{R}_{j+} \mathbf{s}}{\mathbf{s}^H \mathbf{R}_{j-} \mathbf{s}} \quad \text{and} \quad \frac{\mathbf{s}^H \mathbf{R}_{j-} \mathbf{s}}{\mathbf{s}^H \mathbf{R}_{j+} \mathbf{s}}. \quad (5)$$

To solve this optimization problem, the original CSP algorithm includes whitening as the preprocessing step. The idea is geometrically intuitive: If the total power projected on every spatial direction is constrained to be the same, then spatial projections that maximize power for one class will of course maximize the power ratio of this class to the other class. Such spatial projections can easily be found by eigenvalue decomposition of the whitened spatial covariance matrix for each class. A corresponding interpretation of CSP from the semiblind source separation perspective was given in [21].

In effect, mathematically, the objective in the aforementioned optimization problem is recognized as having the form of a generalized Rayleigh quotient. Thus, the optimal spatial projections can be found directly by solving the following generalized eigenvalue problem [14]:

$$\mathbf{R}_{j+} \mathbf{s} = \lambda \mathbf{R}_{j-} \mathbf{s}. \quad (6)$$

The discriminability of each eigenvector is measured by the corresponding eigenvalue. In this case, we can choose the eigenvectors corresponding to the largest $m/2$ (assume m is selected to be even) eigenvalues as the set of spatial projections for class “+”, and the eigenvectors corresponding to the smallest $m/2$ eigenvalues as the set of spatial projections for class “−”, which we denote by $\mathbf{S}_{j+}$ and $\mathbf{S}_{j-}$, respectively. Consequently, if each spatial projection is viewed as a spatial filter, we can get a total of m^2 spatial filters for m temporal filters. To avoid exponential increase of the filter number during iterations, only m spatial filters are selected for further update in each iteration. If we also denote the largest eigenvalue by λ_{j+} and the smallest eigenvalue by λ_{j-} , a viable

way for the selection is to find $j^*+ = \arg \max_{j=1,\dots,m} \lambda_{j+}$ and $j^*- = \arg \min_{j=1,\dots,m} \lambda_{j-}$, then take $\mathbf{S}_{j^*+}$ and $\mathbf{S}_{j^*-}$ as the selected m spatial filters.

2) *Learning Spectral Filters Jointly With the Classifier*: Due to the linearity of spatial and temporal filtering operations, the temporal filters associated with the spatial filters can be improved based on the spatially filtered EEG data. That is to say, contrary to the initial sequence of temporal filtering followed by spatial filtering, we can first apply the already optimized spatial filters to the raw data, and then, update the temporal filters based on the resultant spatially filtered data. Specifying a proper updating rule is critical.

To update temporal filters, we first formulate an alternative description of the feature vector Φ in terms of spectral filters. It is known from signal processing theory that a circulant matrix can be diagonalized by the *discrete Fourier transform* [22]. For a temporal filter $\mathbf{A}_j$, which is circulant, this diagonalization can be posed as

$$\mathbf{F}^H \mathbf{A}_j = \text{diag}([\alpha_j^{(1)}, \dots, \alpha_j^{(T)}]) \mathbf{F}^H \quad (7)$$

where $\mathbf{F} \in \mathbb{R}^{T \times T}$ is the Fourier matrix whose (t, k) th entry is $e^{-\sqrt{-1}2\pi(t-1)(k-1)/T} / \sqrt{T}$ for $1 \leq k, t \leq T$; $\alpha_j^{(i)}$ denotes the eigenvalue corresponding to the i th column of $\mathbf{F}$, or in signal processing terminology, the frequency response of the temporal filter at the i th frequency bin; $\text{diag}([\alpha_j^{(1)}, \dots, \alpha_j^{(T)}])$ is a diagonal matrix with $\alpha_j^{(i)}$ on the i th diagonal entry. It then follows that

$$\begin{aligned} & \mathbf{F}^H \mathbf{A}_j (\mathbf{F}^H \mathbf{A}_j)^H \\ &= \text{diag}([\alpha_j^{(1)}, \dots, \alpha_j^{(T)}]) (\mathbf{F}^H \mathbf{F}) \cdot \\ & \quad \text{diag}^H([\alpha_j^{(1)}, \dots, \alpha_j^{(T)}]) \\ &= \text{diag}([\beta_j^{(1)}, \dots, \beta_j^{(T)}]) = \text{diag}(\boldsymbol{\beta}_j) \end{aligned} \quad (8)$$

where $\beta_j^{(i)} = |\alpha_j^{(i)}|^2$ is the power of the j th temporal filter at the i th frequency bin, which we shall call the spectral coefficient. $\boldsymbol{\beta}_j = [\beta_j^{(1)}, \dots, \beta_j^{(T)}]^H$ denotes the corresponding j th spectral filter formed by the spectral coefficients. The second equality holds because $\mathbf{F}^H \mathbf{F} = \mathbf{I}_T$, where $\mathbf{I}_T$ is the $T \times T$ identity matrix. Equation (1) can now be written as follows by substituting (8) into it

$$\begin{aligned} & \phi_j(\mathbf{X}; \mathbf{A}_j, \mathbf{s}_j) \\ &= \mathbf{s}_j^H \mathbf{X} \mathbf{A}_j \mathbf{A}_j^H \mathbf{X}^H \mathbf{s}_j \\ &= \mathbf{s}_j^H \mathbf{X} \mathbf{F} (\mathbf{F}^H \mathbf{A}_j (\mathbf{F}^H \mathbf{A}_j)^H) \mathbf{F}^H \mathbf{X}^H \mathbf{s}_j \\ &= \mathbf{s}_j^H \hat{\mathbf{X}} \text{diag}(\boldsymbol{\beta}_j) \hat{\mathbf{X}}^H \mathbf{s}_j \\ &= \sum_{i=1}^T \beta_j^{(i)} \mathbf{s}_j^H \hat{\mathbf{x}}^{(i)} \hat{\mathbf{x}}^{(i)H} \mathbf{s}_j \end{aligned} \quad (9)$$

where $\hat{\mathbf{X}} = \mathbf{X} \mathbf{F}$ denotes the discrete Fourier transformed data matrix, and vector $\hat{\mathbf{x}}^{(i)}$ denotes its i th column.

For a temporal filter with real coefficients, its spectral coefficients are symmetric around $T' = \lceil \frac{T}{2} \rceil$ ($\lceil \cdot \rceil$ means rounding to

the nearest integer toward infinity). Combined with the knowledge that the discrete Fourier transform of a real-valued signal is conjugate symmetric around T' [23], it follows that the terms in the summation in the last equality of (9) are conjugate symmetric around T' . Thus, (9) can be expressed by only the first T' frequency bins²

$$\begin{aligned}\phi_j(\mathbf{X}; \beta_j, \mathbf{s}_j) &= 2 \sum_{i=1}^{T'} \beta_j^{(i)} \mathbf{s}_j^H \text{Re}[\hat{\mathbf{x}}^{(i)} \hat{\mathbf{x}}^{(i)H}] \mathbf{s}_j \\ &= \sum_{i=1}^{T'} \beta_j^{(i)} \hat{z}_j^{(i)}\end{aligned}\quad (10)$$

where $\text{Re}[\cdot]$ denotes the real part of $\cdot$, and for any fixed spatial filter $\mathbf{s}_j \in [\mathbf{S}_{j^*+}, \mathbf{S}_{j^*-}]$, we denote the scalar $2\mathbf{s}_j^H \text{Re}[\hat{\mathbf{x}}^{(i)} \hat{\mathbf{x}}^{(i)H}] \mathbf{s}_j$ by $\hat{z}_j^{(i)}$. The variable $\hat{z}_j^{(i)}$ is the power at the i th frequency bin of spatially filtered data.³

From (10), we can see that the band power feature component $\phi_j(\mathbf{X}; \beta_j, \mathbf{s}_j)$ is simply a conic combination of $\hat{z}_j^{(i)}$, $i = 1, \dots, T'$. Therefore, to optimize the temporal filter for a feature component, we can alternatively optimize its corresponding spectral filter. In fact, upon closer examination, it can be seen that the spectral coefficients for all the temporal filters as well as the classifier can be parameterized together. Indeed, substituting (10) for ϕ into (3) yields

$$\begin{aligned}o(\phi) &= \mathbf{w}^H \left[\sum_{i=1}^{T'} \beta_1^{(i)} \hat{z}_1^{(i)}, \dots, \sum_{i=1}^{T'} \beta_m^{(i)} \hat{z}_m^{(i)} \right]^H + b \\ &= \sum_{j=1}^m \sum_{i=1}^{T'} w_j \beta_j^{(i)} \hat{z}_j^{(i)} + b.\end{aligned}\quad (11)$$

Without loss of generality, we assume that the first $m/2$ spatiotemporal filters correspond to the “+” class, and the latter $m/2$ spatiotemporal filters correspond to the “−” class. Then, for the first $m/2$ filters, it is reasonable to assume the following relationship:

$$\left(\sum_{i=1}^{T'} \beta_j^{(i)} \hat{z}_j^{(i)} \right)_+ > \left(\sum_{i=1}^{T'} \beta_j^{(i)} \hat{z}_j^{(i)} \right)_-, \quad j = 1, \dots, \frac{m}{2}\quad (12)$$

where $(\cdot)_k$ denotes the quantity $\cdot$ calculated for a single-trial data of class k . The inequality tells us that the first $m/2$ band power feature components for the “+” class should be larger than those for the “−” class. Similarly, the following inequality

²Note that (10) strictly holds only if $\hat{\mathbf{x}}^{(1)} \hat{\mathbf{x}}^{(1)H} = 0$ when T is odd, while $\hat{\mathbf{x}}^{(1)} \hat{\mathbf{x}}^{(1)H} = 0$ and $\hat{\mathbf{x}}^{(T')} \hat{\mathbf{x}}^{(T')H} = 0$ when T is even. These conditions are satisfied since a bandpass filter is typically applied in the preprocessing step so that the zero ($i = 1$) and the highest ($i = T'$) frequency components are removed.

³By the same token, the spatial covariance matrices in (4) can also be described by the spectral filter: $\mathbf{R}_{j+/-} = \langle \mathbf{X} \mathbf{A}_j \mathbf{A}_j^H \mathbf{X}^H \rangle_{+/-} = 2 \langle \sum_{i=1}^{T'} \beta_j^{(i)} \text{Re}[\hat{\mathbf{x}}^{(i)} \hat{\mathbf{x}}^{(i)H}] \rangle_{+/-}$.

holds for the latter $m/2$ feature components:

$$\left(\sum_{i=1}^{T'} \beta_j^{(i)} \hat{z}_j^{(i)} \right)_+ < \left(\sum_{i=1}^{T'} \beta_j^{(i)} \hat{z}_j^{(i)} \right)_-, \quad j = \frac{m}{2} + 1, \dots, m.\quad (13)$$

In light of (11), in order to maximally discriminate between the two classes, i.e., in order to maximize $|(o(\phi))_+ - (o(\phi))_-|$, the weights of the classifier for the first $m/2$ band power feature components should be opposite in sign to those of the weights for the latter $m/2$ band power feature components, i.e.,

$$\begin{aligned}w_1, \dots, w_{\frac{m}{2}} &\geq 0 \\ w_{\frac{m}{2}+1}, \dots, w_m &\leq 0\end{aligned}\quad (14)$$

where without loss of generality, we assume that the first half of the weights are greater and equal to zero. Combining the constraint $\beta_j^{(i)} \geq 0$ (recall that $\beta_j^{(i)} = |\alpha_j^{(i)}|^2$), we find

$$\begin{aligned}\tilde{w}_j^{(i)} &= w_j \beta_j^{(i)} \geq 0, \quad j = 1, \dots, \frac{m}{2}; \quad i = 1, \dots, T' \\ \tilde{w}_j^{(i)} &= w_j \beta_j^{(i)} \leq 0, \quad j = \frac{m}{2} + 1, \dots, m; \quad i = 1, \dots, T'.\end{aligned}\quad (15)$$

Then, (10) becomes

$$o(\phi) = \tilde{\mathbf{w}}^H \hat{\mathbf{z}} + b\quad (16)$$

where

$$\begin{aligned}\tilde{\mathbf{w}} &= [\tilde{w}_1^{(1)}, \dots, \tilde{w}_1^{(T')}, \dots, \tilde{w}_m^{(1)}, \dots, \tilde{w}_m^{(T')}]^H \\ \hat{\mathbf{z}} &= [\hat{z}_1^{(1)}, \dots, \hat{z}_1^{(T')}, \dots, \hat{z}_m^{(1)}, \dots, \hat{z}_m^{(T')}]^H.\end{aligned}\quad (17)$$

Thus, $\tilde{\mathbf{w}}$ and b parameterize both the spectral filters and the classifier.

From (16), we can see that learning $\tilde{\mathbf{w}}$ and b from the training set $\mathcal{D}$ is equivalent to solving a constrained linear classification problem. However, the problem of learning a hyperplane that separates two sets of data is an ill-posed one, in that there is generally more than one solution to such a problem. A viable way of ensuring the uniqueness of the solution is to adapt an existing well-established linear learning machine according to the constraints in (15). Well-known linear learning machines include *Fisher discriminant analysis* (FDA), *perceptron* of Rosenblatt, *logistic regression*, and *support vector machines* (SVMs) [24]. The SVM is based on the structural risk minimization (SRM) inference principle derived from statistical learning theory [16]. Its idea is to use the maximal margin hyperplane to separate the data. An important characteristic of this type of learning is that its generalization ability is in some sense dimension free, which may avoid the issue of the *curse of dimensionality*. For this reason, SVM can achieve good generalization performance even for very high-dimensional data. Because in our case, $\tilde{\mathbf{w}}$ has a relatively high dimension of mT' compared to the number of available training samples, we decide to adopt the use of maximal margin hyperplane to optimize $\tilde{\mathbf{w}}$. Note that in our case, the maximal margin hyperplane should be constrained by (14).

The question then arises as to whether the constrained maximal margin hyperplane can still generalize well on the unseen

test set. Let Δ denote the margin of the constrained maximal margin hyperplane and R denote the radius of vectors $\hat{\mathbf{z}}$ around the origin, i.e., for any $\hat{\mathbf{z}}$, $\|\hat{\mathbf{z}}\|_2 \leq R$. The following theorem bounds the VC dimension of the set of constrained maximal margin hyperplanes.

Theorem 2.1. The set of maximal margin hyperplanes constrained by (14) has a VC dimension d bounded from above by

$$d \leq \min\left(\left\lceil \frac{R^2}{\Delta^2} \right\rceil, mT'\right) + 1. \quad (18)$$

The proof of this theorem is provided in Appendix I.

Therefore, given the same amount of training data, the generalization ability of the constrained maximal margin hyperplane is at least as good as that of the maximal margin hyperplane.

To calculate the optimal $\tilde{\mathbf{w}}$ and b , consider classifying n trials of training set

$$\{(\hat{\mathbf{z}}_1, y_1), \dots, (\hat{\mathbf{z}}_n, y_n)\} \quad (19)$$

where y_k denotes the label of the k th trial, with $y_k = 1$ when the k th trial belongs to “+” class and $y_k = -1$ otherwise. The optimization can be posed as follows:

$$\begin{aligned} & \min_{\tilde{\mathbf{w}}} \frac{1}{2} \|\tilde{\mathbf{w}}\|_2^2 + C \sum_k \xi_k \\ \text{s.t. } & y_k (\tilde{\mathbf{w}}^H \hat{\mathbf{z}}_k + b) \geq 1 - \xi_k \\ & \xi_k \geq 0, \quad i = 1, \dots, n \\ & \tilde{w}_j \geq 0, \quad j = 1, \dots, \frac{m}{2} \\ & \tilde{w}_j \leq 0, \quad j = \frac{m}{2} + 1, \dots, m. \end{aligned} \quad (20)$$

Here, $\xi_k (k = 1, \dots, n)$ are slack variables introduced to trade-off training errors, which is necessary in the linearly inseparable case. In the current formulation, we adopt the ℓ_1 -norm of $\xi = [\xi_1, \dots, \xi_n]^H$ (also known as *hinge loss*) as the regularization term. The hinge loss is a well motivated proxy for training errors, which itself is nonconvex and NP-hard to optimize. The variable C is the regularization parameter used to control the degree of the tradeoff. An efficient approach for solving (20) to get $\tilde{\mathbf{w}}$ and b is provided in Appendix II.

This completes one iteration, where we obtain m spatial filters contained in $\mathbf{S}_{j^*+}$ and $\mathbf{S}_{j^*-}$, as well as $\tilde{\mathbf{w}}$ and b . The spectral filters on which the subsequent iteration is based can be selected from $\tilde{\mathbf{w}}$. Since the magnitude and the sign of spectral coefficients have no significance, we can simply set the spectral filters to be “normalized” $\tilde{\mathbf{w}}$:

$$\beta_j = \frac{[\tilde{w}_j^{(1)}, \dots, \tilde{w}_j^{(T')}]^T}{\|[\tilde{w}_j^{(1)}, \dots, \tilde{w}_j^{(T')}]^T\|_2}, \quad j = 1, \dots, m. \quad (21)$$

Then, in the same manner, we can reupdate spatial and spectral filters in the subsequent iteration and repeat the procedure.

The initial spectral filters for the first iteration can be set to accommodate a broad frequency range, e.g., 7–30 Hz. The stopping criterion can be chosen by the user, e.g., stopping when the relative change between the eigenvalues λ_{j^*+} or λ_{j^*-} re-

TABLE I
SUMMARY OF THE ISSPL ALGORITHM

Input: the training set $\mathcal{D} = [\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_n]$ corresponding labels $[y_1, y_2, \dots, y_n]$ initial spectral filters $\beta_1, \beta_2, \dots, \beta_m$
For $k = 1$ to n $\hat{\mathbf{X}}_k = \mathbf{X}_k F$ (F is the Fourier matrix whose (t, l) th entry is $e^{-\sqrt{-1}2\pi(t-1)(l-1)/T} / \sqrt{T}$ for $1 \leq l, t \leq T$)
End $T' = \lceil \frac{T}{2} \rceil$
Repeat
For $j = 1$ to m $\mathbf{R}_{j+} \leftarrow 2 < \sum_{i=1}^{T'} \beta_j^{(i)} \text{Re}[\hat{\mathbf{x}}_k^{(i)} \hat{\mathbf{x}}_k^{(i)H}] >_+$ $\mathbf{R}_{j-} \leftarrow 2 < \sum_{i=1}^{T'} \beta_j^{(i)} \text{Re}[\hat{\mathbf{x}}_k^{(i)} \hat{\mathbf{x}}_k^{(i)H}] >_-$ ($\hat{\mathbf{x}}_k^{(i)}$ is the i th column of $\hat{\mathbf{X}}_k$; $\beta_j^{(i)}$ is the i th entry of β_j) calculate $\mathbf{S}_{j+}, \mathbf{S}_{j-}, \lambda_{j+}$, and λ_{j-} via CSP
End $j^*+ \leftarrow \text{argmax}_{j=1,2,\dots,m} \lambda_{j+}$ and $j^*- \leftarrow \text{argmin}_{j=1,2,\dots,m} \lambda_{j-}$
For $j = 1$ to m For $i = 1$ to T' $\hat{\mathbf{z}}_j^{(i)} \leftarrow 2 \mathbf{s}_j^H \text{Re}[\hat{\mathbf{x}}_k^{(i)} \hat{\mathbf{x}}_k^{(i)H}] \mathbf{s}_j$ End $\hat{\mathbf{z}}_k \leftarrow [\hat{\mathbf{z}}_1^{(1)}, \hat{\mathbf{z}}_1^{(2)}, \dots, \hat{\mathbf{z}}_1^{(T')}, \dots, \hat{\mathbf{z}}_m^{(1)}, \hat{\mathbf{z}}_m^{(2)}, \dots, \hat{\mathbf{z}}_m^{(T')}]^H$
End calculate $\tilde{\mathbf{w}}$ and b from $\{(\hat{\mathbf{z}}_k, y_k) k = 1, 2, \dots, n\}$ by solving Equation (20) $\beta_j \leftarrow [\tilde{w}_j^{(1)}, \dots, \tilde{w}_j^{(T')}] / \ [\tilde{w}_j^{(1)}, \dots, \tilde{w}_j^{(T')}]^T\ _2 (j = 1, 2, \dots, m)$
Until stopping criterion satisfied
Return: optimized spatial filters: $\mathbf{S}_{j^*+}$ and $\mathbf{S}_{j^*-}$ optimized spectral filters along with the classifier: $\tilde{\mathbf{w}}$ and b

turned by CSP of two consecutive iterations is less than a preset threshold ε ($0 \leq \varepsilon < 1$) or a maximum number of iterations has been reached. In practice, as has been shown in our own data analysis presented in the Section IV, a satisfactory solution can be obtained after only a few iterations.

Table I summarizes the steps of the ISSPL algorithm.

3) Classifying A Test Sample: To classify a test sample $\mathbf{X}_t \in \mathbb{R}^{D \times T}$, we first calculate $\hat{\mathbf{z}}_t$ as in (10) by applying the optimized spatial filters, then calculate $o(\phi)$ in (16) by applying the optimized $\tilde{\mathbf{w}}$ and b . The sample will be assigned to the “+” class if $o(\phi) > 0$, and to the “−” class otherwise. The flow chart for classifying a test sample is shown in Fig. 1.

C. Comparison With SPECtrally Weighted CSP (SPEC-CSP)

Another iterative algorithm for optimizing spatio-spectral filters proposed recently is SPEC-CSP [13]. Similar to ISSPL, spatial and spectral filters are optimized sequentially in each iteration. To optimize the j th spectral filters corresponding to the “+” class, the following maximization problem is solved:

$$\begin{aligned} & \max_{\beta_j} \frac{\langle \phi_j(\mathbf{X}; \beta_j, \mathbf{s}_j) \rangle_+ - \langle \phi_j(\mathbf{X}; \beta_j, \mathbf{s}_j) \rangle_-}{\sqrt{\text{Var}[\phi_j(\mathbf{X}; \beta_j, \mathbf{s}_j)]_+ + \text{Var}[\phi_j(\mathbf{X}; \beta_j, \mathbf{s}_j)]_-}} \\ \text{s.t. } & \beta_j^i \geq 0, \quad i = 1, 2, \dots, T' \end{aligned} \quad (22)$$

where $\phi_j(\mathbf{X}; \beta_j, \mathbf{s}_j)$ is defined in (10). The spectral filters corresponding to the “−” class can be optimized in the same way

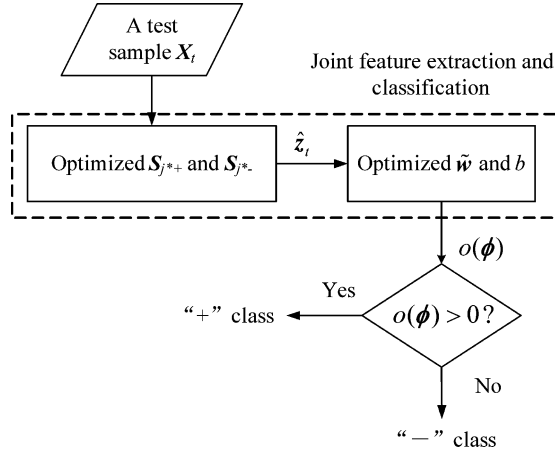


Fig. 1. Proposed classification procedure for a test sample.

by simply exchanging the labels in (22). One may find that (22) is essentially the square root of the Rayleigh quotient used in FDA, with additional constraints. However, the solution cannot be easily found unless simplifying assumptions are made. In [13], the authors made an assumption that the signal is a stationary Gaussian process so that the frequency components were independent from each other for a given class label. In this case, a concise solution can be found by solving the dual problem of (22).

There are several weaknesses with SPEC-CSP that are as follows.

- 1) The assumption of independence is untenable from the neurophysiological perspective. For instance, at least some β rhythms are believed to be harmonics of μ rhythms [25].
- 2) Spectral filters for different feature components are optimized individually in the form of (22), which is essentially a feature extraction method that aids in classification without specifically aiming at solving the classification problem. The objective is built upon the intuition that “separation” (in the statistical sense, as characterized by first and second moments) is useful. To achieve high classification accuracy, it is preferable to optimize all spectral filters with a common objective for classification.
- 3) Maximizing the square root of the Rayleigh quotient as in (22) might not necessarily guarantee good generalization performance of the optimized spectral filters. It is still theoretically an open issue as to the generalization error bound of FDA. Evidently, SPEC-CSP is faced with the same dilemma.

In contrast, our proposed new approach does not make any additional statistical assumption to carry out the optimization, and all spectral filters are optimized simultaneously under a common optimization framework in which the classifier is also implicitly learned. In this way, our approach preserves the intuition of finding a direction that separates (in the maximal margin sense) the classes, while aimed directly at solving the classification problem. Moreover, as has been demonstrated earlier, the idea of the maximal margin hyperplane can ensure good generalization ability of the learned spectral filters and the classifier.

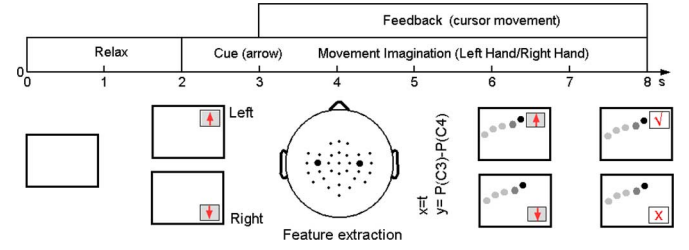


Fig. 2. Paradigm of the online motor imagery experiment. During the first 2 s, while the screen was blank, the subject was in the relax state. Immediately after that, a visual cue (an arrow) was presented on the screen, indicating the imagery task to be performed. The arrow pointing upward and downward indicated the tasks of imagination of the left-hand and the right-hand movements respectively. At the third second, a cursor started to move horizontally in a constant speed from the left side to the right side of the screen. The horizontal position of the cursor is the time point in the current trial; the vertical position of the cursor is the accumulated power difference between C3 and C4. At the end of the trial, a true or false mark appeared to indicate the classification result of the trial and the subject was asked to relax and wait for the next task.

III. EXPERIMENTAL EVALUATION

The feature extraction and classification abilities of the ISSPL algorithm were tested on two EEG datasets. The first dataset was recorded from our own online motor imagery experiments, where we compared the performance of ISSPL against wideband CSP and SPEC-CSP. The objective is to see if ISSPL can surpass these state-of-the-art approaches in classification performance. The second EEG dataset we used was dataset IVa from BCI Competition III, where we benchmarked ISSPL against wideband CSP, CSP with temporal filters tuned according to the winning entry of the dataset, and SPEC-CSP. The objective is to confirm if ISSPL can indeed automatically find spatio-spectral filters with performance comparable to intensively tuned ones.

A. Experiment I: Dataset From Our Own Motor Imagery Experiments

1) *Description*: Six healthy volunteers (three males and three females; right handed; 21–24 years old) participated in our online BCI control experiments with visual feedback. The left-hand and right-hand movement imagination were designated to control the vertical movement of a cursor. Fig. 2 shows the paradigm of the feedback experiment. The EEG recording was made using a BioSemi ActiveTwo system. According to the 10/20 international system, 32 channels were placed at positions including C3/C4 and FCz electrodes over the primary motor area (M1) and the supplementary motor area (SMA). All the channels were referenced to the left earlobe. Signals were digitized at 256 Hz. For each subject, a dataset comprised of 70 trials (35 trials per class) recorded in a single session was used for off-line algorithmic studies.

A time period of 2 s (between 2.5 and 4.5 s) was chosen from each trial for the comparison of ISSPL with wideband CSP and SPEC-CSP. For wideband CSP, a bandpass temporal filter from 7 to 30 Hz (*wideband*) was applied to the data to restrict the analysis to μ and β rhythms. For SPEC-CSP and ISSPL, the initial spectral filters were also set to be flat within 7–30 Hz, and only spectral coefficients within this passband

TABLE II
CLASSIFICATION ACCURACIES $\pm$ STANDARD DEVIATION (%) OF WIDEBAND CSP, THEORETICAL/MODIFIED SPEC-CSP, AND ISSPL

Subject	Wide-Band CSP + SVM	Theoretical SPEC-CSP + SVM	Modified SPEC-CSP + SVM	ISSPL
MY	93.50 $\pm$ 3.11	90.75 $\pm$ 1.69	97.00 $\pm$ 2.58	98.67 $\pm$ 1.12
WW	96.42 $\pm$ 1.31	95.33 $\pm$ 1.72	95.17 $\pm$ 1.88	97.50 $\pm$ 1.42
SJH	93.67 $\pm$ 1.48	92.25 $\pm$ 2.52	93.92 $\pm$ 2.39	95.08 $\pm$ 2.10
ZYJ	94.75 $\pm$ 1.62	95.92 $\pm$ 1.86	95.42 $\pm$ 1.93	96.75 $\pm$ 1.69
FL	73.42 $\pm$ 5.16	78.25 $\pm$ 6.79	78.48 $\pm$ 8.44	80.00 $\pm$ 5.10
HLW	70.07 $\pm$ 3.72	74.00 $\pm$ 7.74	78.27 $\pm$ 3.83	83.33 $\pm$ 2.76

were considered for further optimization across iterations, while spectral coefficients outside 7–30 Hz were consistently set to zero. In particular, two cases were considered separately for SPEC-CSP: the one with the theoretical spectral filters obtained by solving (22) (referred to below as *theoretical* SPEC-CSP) and the one with the additional prior filters included (referred to below as *modified* SPEC-CSP).⁴ Classifications for feature vectors extracted by wideband CSP and theoretical/modified SPEC-CSP were carried out using SVM with a linear kernel. Both the regularization parameter and the classification accuracies were obtained using 10×10 fold cross-validation. For simplicity, only one spatial filter per class for wideband CSP, and one set of spatiotemporal filters per class for theoretical/modified SPEC-CSP and ISSPL were used (i.e., $m = 2$). The number of iterations for theoretical/modified SPEC-CSP and ISSPL was set to 5.

2) *Results*: The results are shown in Table II, in which the best classification accuracy for each subject is highlighted in bold. As can be seen, ISSPL and modified SPEC-CSP in general outperform wideband CSP. ISSPL in particular consistently performs the best for all the six subjects, especially for FL and HLW, whose performance are relatively poor. It is also interesting to note that theoretical SPEC-CSP performs even worse than wideband CSP in most cases, which is consistent with the findings in [13]. This might be due to the overfitting caused by high dimensionality involved in the optimization of spectral filters and the inaccuracy of the statistical assumption that has been made. Incorporation of the prior filter can alleviate the issue to a certain extent. However, even in such a case, ISSPL still works better. The reason behind this is that although the prior filter is regarded as containing prior information as its name suggests, in effect, it still has to be obtained from the data, not information known in advance. The learning goal is always to achieve good generalization performance. Although ISSPL does not explicitly incorporate the prior filters, the fact that its optimization of spectral filters is directly related to the generalization perfor-

⁴The j th prior filter encodes the knowledge on the spectrum of the j th spatially filtered signal, i.e., $\hat{z}_j^{(i)}$ ($i = 1, \dots, T'$) as in (10). It is defined as $(\langle \hat{z}_j^{(i)} \rangle_+ + \langle \hat{z}_j^{(i)} \rangle_-)/2$ ($i = 1, \dots, T'$). The j th spectral filter for modified SPEC-CSP is calculated by multiplying the j th prior filter with the j th theoretical spectral filter at each frequency bin.

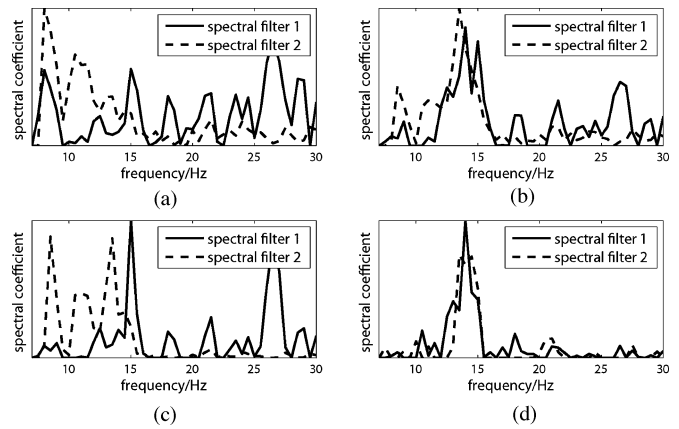


Fig. 3. Spectral filters for subject MY. Spectral filter 1 and 2 correspond to left-hand movement imagery and right-hand movement imagery, respectively. (a) Theoretical spectral filters learned by theoretical SPEC-CSP. (b) Prior filters (one for each spectral filter). (c) Spectral filters learned by modified SPEC-CSP. (d) Spectral filters learned by ISSPL.

mance enables it to learn the discriminative information in the prior filters from the data.

We take subject MY as an example to illustrate this point. The theoretical spectral filters learned by theoretical SPEC-CSP are shown in Fig. 3(a). Note that spectral filter 2 exhibits an undesirably large variability across frequency bins. The prior filters (one for each spectral filter) are shown in Fig. 3(b). Note that these have a large peak around 12–16 Hz. If the prior filters are incorporated into SPEC-CSP, the resultant spectral filters are shown in Fig. 3(c). Fig. 3(d) shows the spectral filters learned by ISSPL. It can be seen that ISSPL detects the large peak in the prior filters, which indicates that the nearby frequency range is useful for discrimination of the two classes. The classification results of subject MY also indicate that this type of learning is superior to explicitly incorporating the prior filters in the spectral filters.

To show the variation of ISSPL's performance across iterations, we plot its classification accuracies for subject MY within five iteration steps in Fig. 4. Generally, it can be observed that there is significant improvement in performance after the first iteration. In the following iterations, the performance changes slowly, sometimes exhibiting a small amount of degradation. In fact, when we performed the cross-validation for each subject, we found that if a stopping criterion was chosen in the way as

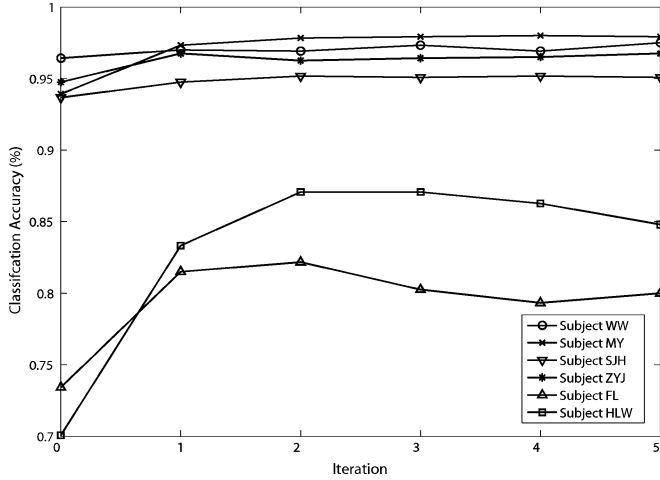


Fig. 4. Classification accuracies (%) across iterations for ISSPL on different subjects. All classification accuracies are obtained by 10×10 cross-validation. The zero bin on the horizontal axis corresponds to the wideband CSP.

has been suggested at the end of Section II-B2, where ε was set to be 0.05, the algorithm usually converged in less than three steps. This is quite consistent with Fig. 4, where the highest classification accuracies are attained at around step 3.

B. Experiment II: Dataset IVa in BCI Competition III

1) *Description*: The EEG data were recorded from five healthy subjects (aa, al, av, aw, and ay). According to the international 10/20 system, 118 channels were placed. Signals were downsampled to 100 Hz for analysis. During each trial, the subject was required to perform either of two motor imagery tasks for 3.5 s: right-hand and right-foot movements. A total of 140 trials were collected for each subject and each task.

ISSPL was benchmarked against CSP with manually adjusted temporal filters. For CSP, all relevant parameters, including the temporal filter, channels, time window, and the number of spatial filters, were tuned according to the winning entry of the dataset [26], which happens to be from the authors' laboratory. For a fair comparison, the same parameter settings were used by ISSPL except for spectral filters that had to be optimized. The number of iterations was fixed to 3 in light of the results in Experiment I. To show the extent to which ISSPL can boost the performance of wideband CSP (7–30 Hz), we also present the classification accuracies of wideband CSP and modified SPEC-CSP on this dataset. Given the disappointing classification performance of theoretical SPEC-CSP in Experiment I, we excluded it from further comparison.

2) *Results*: The results obtained by 10×10 cross-validation are shown in Table III, in which the best classification accuracy for each subject is again highlighted in bold.

As can be seen, encouraging results are obtained. ISSPL is the best all-round performer, producing the best classification rates in four out of five subjects. For subject aa, CSP with manually tuned temporal filters is slightly better than ISSPL. But it is important to note that the parameters for the winning entry were tuned intensively. With no surprise, CSP performs rather poorly if a wideband temporal filter is adopted.

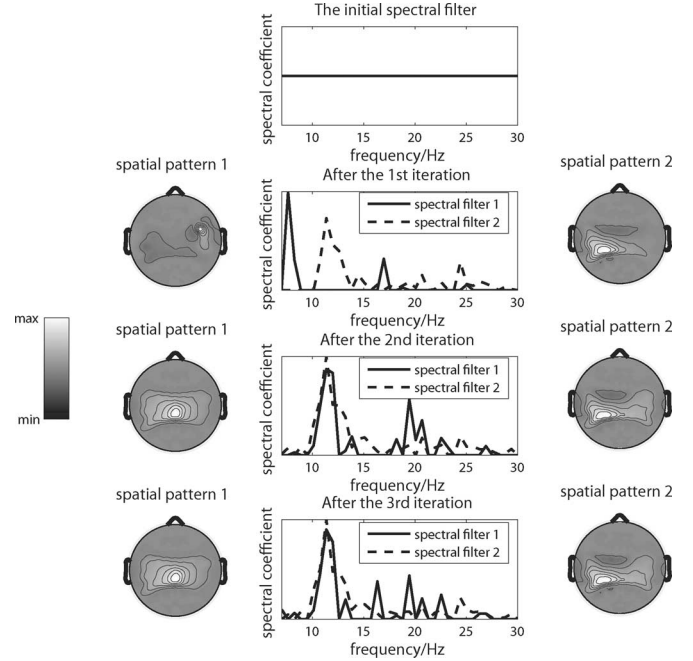


Fig. 5. Updating process of the spatio-spectral filters for subject aw. Spatial pattern 1 and spectral filter 1 correspond to right-hand movement imagery. Spatial pattern 2 and spectral filter 2 correspond to right-foot movement imagery. In the second iteration, the spatial pattern of right-hand movement imagery shifts to the characteristic ERD location for right-foot movement imagery. The peak of the corresponding spectral filter is updated from 7 to 8 Hz in the first iteration to 10–13 Hz in the second iteration based on the optimized spatial filters.

We next present a concrete example to see how the spatio-spectral patterns are updated across iterations. Fig. 5 shows such an updating process for subject aw. In the second iteration, the spatial pattern of left-hand movement imagery shifts to the mid-central foot representation area. This is desirable since in that area, the band power of the activity when imagining left-hand movement would be larger than that of the activity when imagining foot movement. The peak of the corresponding spectral filter is updated from 7 to 8 Hz in the first iteration to 10–13 Hz in the second iteration based on the optimized spatial filters. Note there is no conspicuous change of the spatio-spectral patterns in the third iteration. This example reveals that ISSPL is effective in finding the subject-specific optimal spatio-spectral patterns in an iterative way.

IV. DISCUSSION AND CONCLUSION

A. Theoretical Issues

In each iteration of ISSPL, CSP is employed for optimizing spatial filters, while the idea of maximal margin hyperplane is adopted for optimizing spectral filters and the classifier. Since CSP and maximal margin hyperplane are very different in their optimization objectives, it is difficult to prove the convergence of ISSPL in theory. However, a rule of thumb we have observed empirically in the preceding experiments is that only a few iterations are required for ISSPL to achieve fairly good classification performance. A nice property of ISSPL we would like to stress is that in each iteration, given spatial filters, the joint optimization

TABLE III
CLASSIFICATION ACCURACIES $\pm$ STANDARD DEVIATION (%) OF WIDEBAND CSP, CSP WITH MANUALLY TUNED TEMPORAL FILTERS, MODIFIED CSP, AND ISSPL

Subject	Wide-Band CSP + SVM	CSP with manually tuned temporal filters + SVM	Modified SPEC-CSP + SVM	ISSPL
aa	72.50 $\pm$ 1.04	93.93 $\pm$ 3.78	86.79 $\pm$ 6.31	93.57 $\pm$ 3.28
al	89.29 $\pm$ 4.32	99.29 $\pm$ 1.51	97.50 $\pm$ 2.94	100 $\pm$ 0.00
av	75.00 $\pm$ 4.76	74.64 $\pm$ 6.40	76.79 $\pm$ 9.41	79.29 $\pm$ 7.10
aw	87.50 $\pm$ 1.06	98.93 $\pm$ 1.73	93.93 $\pm$ 3.78	99.64 $\pm$ 1.13
ay	95.36 $\pm$ 2.41	96.00 $\pm$ 3.58	92.50 $\pm$ 3.93	98.57 $\pm$ 3.01

of spectral filters and the classifier is formulated as a convex quadratic programming problem, for which a global optimum is guaranteed to be attained. Moreover, it is more convenient and straightforward to optimize spectral filters in the frequency domain than to optimize FIR filters in the temporal domain as adopted by CSSSP [12], since the issue of determining the filter's order is completely avoided.

B. Implementational Issues

In [13], Tomika *et al.* suggested using the Welch method to estimate the power spectrum in place of $\hat{z}_j^{(i)}$ in (10). Yet, the computational complexity would greatly increase in that case since not only spectra of single channels but also cross-spectra between any two channels should be estimated, and extra effort is needed to manually adjust the associated parameters such as the window type and size. Hence, we simply used the periodogram method by multiplying the discrete Fourier transform as in (10).

The computational complexity of solving (20) is comparable to the quadratic program that one has to solve for SVM, depending on the size of the training set and the dimension of the spectral filters. While we have used the conjugate gradient method to solve (20), we believe that there is much to gain from exploiting analogies to the SVM and developing specialized optimization procedures for our algorithm, in particular procedures that break the data into subsets [27], which would potentially result in a faster online speed.

C. Further Applications

There can be several further potential applications and extensions of ISSPL.

- 1) The subprocedure for optimizing spectral filters in ISSPL can be applied to simple two-channel motor imagery BCI systems that only use C3 and C4 channels. There is no need for iteration in that case, and one only needs to remove the spatial filtering subprocedure and replace the spatially filtered signals by signals recorded from C3 and C4 channels.
- 2) The ISSPL algorithm can be extended to perform multiclassifications by pairwise combination. Specifically, the original multiclassification problem can be broken into several binary classification subproblems. We can perform

the classification by ISSPL for each subproblem, and then obtain the final result by voting.

- 3) The same methodology in ISSPL can also be employed for determining time windows in the current motor imagery BCI system. The problem of selecting appropriate time points for classification is equivalent to finding an optimal filter in the temporal domain with the diagonality property, and can, therefore, be reduced to a conic combination form in the same fashion as the optimization of spectral filters in ISSPL. In this case, the idea of maximal margin hyperplane can be used to solve the problem.
- 4) Although the new framework adopted in ISSPL is employed to extract the band power feature, similar effort can be attempted to design spatio-spectral filters for other useful features, e.g., phase synchronization between different brain regions [28]. Although in that case, the success will greatly depend on the tractability of the formulated optimization problems.

D. Concluding Remarks

In this paper, we have presented the ISSPL algorithm for classifying single-trial EEG data during motor imagery. In the adopted framework, feature extraction and classification are no longer treated as independent stages: spectral filters and the classifier are parameterized jointly in the maximal margin hyperplane for optimization, and thereby their generalization performance can be controlled.

Results on two off-line data experiments demonstrated the efficacy of ISSPL. The resultant spectral filters did not suffer from the potential overfitting problem typical of learning high-dimensional features, and only a few steps of iterations were needed to obtain satisfactory classification performance. The algorithm is also applicable for online experiment because of its low computational complexity, but further work is necessary before it can be used for a practical BCI.

The Matlab code of the ISSPL algorithm is available upon request.

APPENDIX I

PROOF OF THEOREM 2.1

The VC dimension of a set of functions is defined as the maximum number of vectors that can be shattered by this set of functions. Since the set of maximal margin hyperplanes

constrained by (13) is just a subset of the set of maximal margin hyperplanes, it should have less freedom. Thus, the maximum number of vectors that can be shattered by it clearly does not exceed what can be shattered by the set of maximal margin hyperplanes, and it follows that the set of constrained maximal margin hyperplanes has a VC dimension no greater than that of the set of maximal margin hyperplanes. It has already been demonstrated in [16] that the VC dimension of a set of maximal margin hyperplanes is bounded from above by $(\lceil \frac{R^2}{\Delta^2} \rceil, mT') + 1$. Then, it follows directly that the inequality in the theorem holds. ■

APPENDIX II

AN APPROACH FOR SOLVING (20)

Equation (20) is a convex quadratic programming problem and can be directly solved. However, we notice that for this problem strong duality holds, i.e., the optimal duality gap is zero [29]. Duality theory provides a convenient and efficient way to deal with the constraints, and slack variables will vanish from the dual problem. Moreover, solving dual problem might be of use when one needs a criterion to tell the importance of each sample. Thus, we would like to equivalently solve the Lagrange dual problem. The Lagrangian of the primal problem is

$$\begin{aligned} L = & \frac{1}{2} \|\tilde{\mathbf{w}}\|^2 - \sum_{k=1}^n \nu_k y_k (\tilde{\mathbf{w}}^H \hat{\mathbf{z}}_k + b) + \sum_{k=1}^n \nu_k \\ & - \sum_{k=1}^n \nu_k \xi_k + C \sum_{k=1}^n \xi_k - \sum_{j=1}^{m/2} \rho_j \tilde{w}_j \\ & + \sum_{j=m/2+1}^m \rho_j \tilde{w}_j - \sum_{k=1}^n \eta_k \xi_k \end{aligned} \quad (23)$$

where ν_k , ρ_j , η_k ($k = 1, \dots, n$; $j = 1, \dots, m$) are Lagrange multipliers.

To get the Lagrange dual function, we must first minimize L with respect to primal variables $\tilde{\mathbf{w}}$, b , and ξ . Fermat theorem implies that the gradient of L with respect to $\tilde{\mathbf{w}}$, b , and ξ should vanish at the minimum, which gives the conditions

$$\begin{aligned} \tilde{\mathbf{w}} = & \sum_{k=1}^n \nu_k y_k \hat{\mathbf{z}}_k \\ & + [\rho_1, \dots, \rho_{m/2}, -\rho_{m/2+n}, \dots, -\rho_m] \\ \sum_{k=1}^n \nu_k y_k = & 0 \\ -\nu_k + C - \eta_k = & 0, \quad k = 1, \dots, n. \end{aligned} \quad (24)$$

We can then substitute these conditions into Lagrangian L , and after mathematical manipulations, we arrive at the Lagrange dual function

$$\begin{aligned} g = & \underbrace{[1, 1, \dots, 1, 0, 0, \dots, 0]}_n \gamma \\ & - \frac{1}{2} \gamma^H (\overline{\mathbf{Q}}_1 + \overline{\mathbf{Q}}_2 + \overline{\mathbf{Q}}_3) \gamma \end{aligned} \quad (25)$$

where

$$\begin{aligned} \gamma = & (\gamma_l)_{1 \leq l \leq m+n} \\ = & [\nu_1, \dots, \nu_n, \rho_1, \dots, \rho_{m/2}, -\rho_{m/2+1}, \dots, -\rho_m]^H \\ \mathbf{Q}_1 = & ([\hat{\mathbf{z}}_1, \hat{\mathbf{z}}_2, \dots, \hat{\mathbf{z}}_n]^H [\hat{\mathbf{z}}_1, \hat{\mathbf{z}}_2, \dots, \hat{\mathbf{z}}_n]) \odot (\mathbf{y} \mathbf{y}^H) \\ \overline{\mathbf{Q}}_1 = & \begin{pmatrix} \mathbf{Q}_1 & 0_{n \times m} \\ 0_{m \times n} & 0_{m \times m} \end{pmatrix}, \quad \mathbf{Q}_2 = \begin{pmatrix} y_1 \hat{\mathbf{z}}_1^H \\ y_2 \hat{\mathbf{z}}_2^H \\ \vdots \\ y_n \hat{\mathbf{z}}_n^H \end{pmatrix} \\ \overline{\mathbf{Q}}_2 = & \begin{pmatrix} 0_{n \times n} & \mathbf{Q}_2 \\ \mathbf{Q}_2^H & 0_{m \times m} \end{pmatrix} \\ \overline{\mathbf{Q}}_3 = & \begin{pmatrix} 0_{n \times n} & 0_{n \times m} \\ 0_{m \times n} & \mathbf{I}_m \end{pmatrix}. \end{aligned} \quad (26)$$

Here, $\mathbf{I}_m$ is the $m \times m$ identity matrix, while $0_{m \times n}$ represents an $m \times n$ matrix, all of whose elements are zero; $\mathbf{y} = [y_1, \dots, y_n]^T$; $\odot$ means element-wise product of two matrices. To maximize g with constraints $0 \leq \gamma_l \leq C$ ($l = 1, \dots, n$), $\gamma_l \geq 0$ ($l = n+1, \dots, n+m/2$), $\gamma_l \leq 0$ ($l = n+m/2+1, \dots, n+m$), and $\sum_{l=1}^n \gamma_l y_l = 0$ is again a quadratic optimization problem, which can be efficiently solved by conjugate gradient method. After that, the optimal coefficients $\tilde{\mathbf{w}}$ can be obtained as $\sum_{l=1}^n \gamma_l y_l \hat{\mathbf{z}}_l + [\gamma_{n+1}, \dots, \gamma_{n+m}]^H$. The intercept b can be easily found by using the Karush–Kuhn–Tucker complementary condition

$$\gamma_l (y_l (\tilde{\mathbf{w}}^H \hat{\mathbf{z}}_l + b) - 1) = 0, \quad l = 1, \dots, m \quad (27)$$

by computing b for all l with $\gamma_l \neq 0$ and taking their mean value.

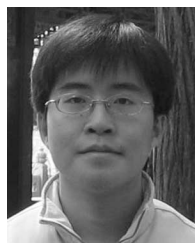
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